DWH Session 1

1. What is the difference between OLTP and OLAP systems? Provide examples of each.

Feature	OLTP	OLAP
Purpose	Run daily operations	Data analysis and decision making
Data Type	Current operational data	Historical aggregated data
Query Type	Simple transactions	Complex analytical queries
Operations	INSERT, UPDATE, DELETE	Mostly SELECT
Users	Clerks, customers, applications	Managers, analysts
Database Design	Highly normalized	Denormalized (Star schema, cubes)
Example	ATM system, order processing	Data warehouse, sales analysis

2. Explain the ETL process and its three main steps.

The **ETL process** is a fundamental concept in **data warehousing** and business intelligence systems. The term **ETL** stands for **Extract, Transform, and Load**, which are the three main steps used to collect data from different sources, prepare it for analysis, and store it in a data warehouse.

1. Extract :
   In this stage, data is collected from different sources such as databases, spreadsheets, APIs, or even flat files like CSV files. These sources may belong to different systems like sales systems, inventory systems, or customer management systems. The goal of extraction is simply to gather the required data without modifying it.

2. Transform :
   This stage is where the extracted data is **cleaned, organized, and prepared** so that it can be used for analysis. During transformation, several operations may occur such as

   - removing duplicate records
   - correcting inconsistent data formats
   - filtering unnecessary data
   - combining data from multiple sources
   - calculating new values

3. Load :
   In this stage, the transformed data is **stored in a destination system**, usually a **data warehouse** or a **data mart**. Once the data is loaded, it becomes available for analytical tools, reports, and dashboards.

3. What are the key characteristics of a Data Warehouse?

1. Subject-Oriented:
A data warehouse organizes data around major business subjects such as customers, products, sales, or employees rather than daily operations.
This organization helps analysts easily study and understand specific areas of the business.

2. Integrated:
A data warehouse combines data from multiple sources such as databases, spreadsheets, and enterprise systems.
During integration, the data is cleaned and standardized so that it follows the same format and naming conventions.

3. Time-Variant:
A data warehouse stores historical data over long periods of time, such as months or years.
This allows organizations to analyze trends, compare past and present performance, and make long-term decisions.

4. Non-Volatile:
Once data is loaded into the data warehouse, it is not frequently modified or deleted**.**
Instead, new data is added periodically, which keeps the stored data stable for consistent analysis and reporting.

4. What is the difference between Normalization and Denormalization, and when would you use each?

Normalization:
Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity. This is done by dividing data into multiple related tables and linking them using keys. The main goal of normalization is to ensure that each piece of information is stored only once, which prevents inconsistencies and makes updates easier.

Normalization is commonly used in transactional systems (OLTP) such as banking systems, e-commerce platforms, and inventory management systems.

Denormalization:
Denormalization is the process of combining tables or adding redundant data to reduce the number of joins needed in queries. Instead of separating data into many small tables, some data is intentionally duplicated to make queries faster and simpler. The main goal of denormalization is to improve query performance, especially when dealing with large amounts of data.

Denormalization is commonly used in data warehouses and analytical systems (OLAP) where the focus is on reading and analyzing data rather than updating it.

5. Describe the differences between Structured, Semi-Structured, and Unstructured Data with real-world examples.

Structured Data :
Structured data is data that is organized in a clear and predefined format, usually in rows and columns like tables in a relational database.
Because the structure is fixed, it is easy to store, search, and analyze using tools such as SQL.
This type of data follows a strict schema where each field has a specific data type and meaning.

Examples :

- **Banking database** where customer information is stored in tables with columns such as CustomerID, Name, AccountBalance, and AccountType.
- **E-commerce order table** that stores order ID, product ID, price, and purchase date.

Semi-Structured Data :
Semi-structured data does not follow a strict table structure like structured data, but it still contains some organizational elements such as

- Tags
- Keys
- markers that help identify different pieces of information. This makes it more flexible than structured data while still allowing systems to interpret it.

Common Examples :

- JSON files
- XML files

Unstructured Data :
Unstructured data is data that does not have a predefined format or structure, making it more difficult to store and analyze using traditional database systems.

Examples :

- Photos
- Videos
- audio files
- PDF files

6. What are the different types of data loading techniques in a Data Warehouse?

Full Load:
Full load means that all data from the source system is loaded into the data warehouse every time the ETL process runs. In this method, the existing data in the destination table is usually cleared and replaced with the new data. This approach is simple to implement because it does not require tracking which records have changed.

Incremental Load :
Incremental loading means that only the new or modified data since the last update is loaded into the data warehouse. Instead of copying the entire dataset again, the ETL process identifies changes using timestamps, change tracking, or unique identifiers.

Real-Time Load:
Real-time loading allows data to be transferred to the data warehouse immediately or with very minimal delay after it is generated in the source system.

7. What is the difference between a Data Warehouse, Data Lake, and Data Lakehouse?

Feature	Data Warehouse	Data Lake	Data Lakehouse
Data Type	Mainly structured data	Structured, semi-structured, and unstructured data	Structured, semi-structured, and unstructured data
Data State	Data is cleaned and processed before storage	Data is stored in its raw format	Can store raw data but also support processed data
Schema Approach	Schema is defined before storing data (Schema-on-Write)	Schema is applied when the data is used (Schema-on-Read)	Supports both schema-on-write and schema-on-read
Purpose	Business intelligence, reporting, and analytics	Big data storage, exploration, and machine learning	Combines analytics performance with flexible storage
Users	Business analysts and managers	Data scientists and engineers	Both analysts and data scientists
Performance for Queries	Optimized for fast analytical queries	Not optimized for complex analytical queries	Provides improved analytical performance similar to data warehouses
Examples of Data	Sales reports, customer transactions, financial records	Logs, videos, images, IoT sensor data, social media data	Combination of warehouse data and large raw datasets

8. What is a Data Mart and why is it used?

A Data Mart is a subset of a data warehouse that is focused on a specific business area, department, or subject, such as sales, marketing, or finance. Unlike a full data warehouse that contains all organizational data, a data mart stores only the relevant data needed by a particular group of users. This makes it faster and easier for teams to access and analyze the information they need.
The main reason a data mart is used is to improve performance and simplify access for users who only need a limited scope of data.

9. What is a Data Warehouse (DWH)? How does it differ from a traditional database?

Data Warehouse:
A Data Warehouse is a system designed to store and analyze large volumes of historical data from multiple sources. It is optimized for reporting, business intelligence, and decision-making, rather than daily operations.

Differences from a traditional database:
Purpose: Traditional databases (OLTP) handle daily transactions; data warehouses (OLAP) support analysis and reporting.
Data: Databases store current operational data; warehouses store historical and aggregated data.
Design: Databases are highly normalized; warehouses are often denormalized for faster queries.
Queries: Databases handle simple, short transactions; warehouses handle complex, long-running analytical queries.

10. What are the different types of databases (Relational/SQL-based vs. Non-Relational/NoSQL-based)? Give examples[i]

1. Relational Databases (SQL-based)
Relational databases store data in tables with rows and columns, and the relationships between tables are defined using keys. They follow a fixed schema and use SQL (Structured Query Language) for querying and managing data.

Characteristics:
Structured data with a predefined schema
Supports transactions and ensures data integrity (ACID properties)
Good for applications that require consistent and organized data

Examples:
MySQL – widely used for web applications
Microsoft SQL Server – used in enterprise systems
Oracle Database – popular for large-scale business applications

2. Non-Relational Databases (NoSQL-based)
NoSQL databases are designed to handle unstructured or semi-structured data and are more flexible in terms of schema. They are optimized for scalability, high performance, and handling large volumes of data.

Types of NoSQL databases:
Document-based: Stores data as JSON-like documents (e.g., MongoDB)
Key-Value: Stores data as key-value pairs (e.g., Redis)

Characteristics:
Flexible schema, can store varied data types
Scales easily across multiple servers
Often used in big data, real-time apps, and web-scale systems